

Rational Expressions & Equations — Test Redo Mix

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Standard: CCSS.MATH.CONTENT.HSA-APR.D.6 & HSA-REI.A.2 — Rewrite rational expressions; solve rational equations and identify extraneous solutions.

△ READ FIRST — Two Easy-to-Miss Ideas

1. SIMPLIFY vs SOLVE: *Simplify* = rewrite an expression (no equals sign → answer is an expression; assume denom $\neq 0$). *Solve* = find x in an equation (equals sign → answer is a number; check extraneous).

2. OPPOSITES: $(2 - x) = -(x - 2)$, $(3 - x) = -(x - 3)$, $(y - x) = -(x - y)$. Whenever you see "flipped" binomials, factor out -1 from one side so the other cancels.

Part A — SIMPLIFY (6 problems, no denominator = 0)

Directions: *Simplify each expression. Show factoring & cancellation. State restrictions where asked. Do not solve — there is nothing to solve.*

A-1 · Linear Simplify (cancels cleanly to a linear or constant expression)

1. Simplify **and state the restriction:** $\frac{a - b}{a^2 - b^2}$

Hint: factor denominator as a difference of squares. Write answer and restriction like $\frac{1}{\dots}$, $a \neq b$.

2. Simplify (opposites): $\frac{y}{x - y} + \frac{x}{y - x}$

Hint: $(y - x) = -(x - y)$.

3. Simplify (opposites): $\frac{3}{x - 3} - \frac{x}{3 - x}$

A-2 · Quadratic Simplify (factor trinomials, difference of squares, division)

4. Simplify: $\frac{9m^2 - 4}{3m^2 + 4m - 4}$

5. Simplify: $\frac{x^2 - 4}{x^2 - x - 6} \div \frac{x^2 + 4x + 4}{x^2 - 9}$

Hint: division $\rightarrow$ multiply by reciprocal, then factor everything.

6. Simplify (bonus-style): $\left(y - \frac{2}{y-1}\right) \left(y + \frac{1}{y-2}\right)$

Part B — SOLVE (6 problems, check extraneous)

Directions: Solve each equation. Always **check** candidate solutions against the original denominators. Reject any that make a denominator zero (extraneous).

B-1 · Linear Solve (reduces to linear after clearing denominators)

7. Solve: $\frac{3}{x+2} = \frac{1}{x-2}$

Hint: cross-multiply.

8. Solve: $\frac{x-1}{x+3} = 2$

9. Solve: $\frac{2}{x-1} + \frac{3}{x-1} = \frac{5}{2}$

B-2 · Quadratic Solve (reduces to a quadratic — extraneous possible)

10. Solve: $\frac{x}{x-1} - \frac{2}{x+1} = \frac{4}{x^2-1}$

11. Solve: $\frac{1}{x} + \frac{1}{x+3} = \frac{1}{2}$

12. Solve: $\frac{2x}{x-3} = \frac{6}{x-3} + \frac{x}{2}$